

Classical Linear Codes

$$\Sigma \in \{0,1\}^n \quad C \subseteq \Sigma^n \quad u, v \in C \rightarrow u \oplus v \in C$$

$$u \oplus v = u + v \text{ modulo } 2$$

$\perp$
 $0^n \in C$ for all
classical linear codes

$\rightarrow C_n \rightarrow$ 3-bit repetition

$\rightarrow (7,4,3)$ - Hamming code

→ Describing Classical Linear Codes

→ Generators

Minimal list of strings $u_1, \dots, u_m \in \Sigma^n$ such that

$$C = \left\{ \alpha_1 u_1 + \dots + \alpha_m u_m : \alpha_1, \dots, \alpha_m \in \{0, 1\} \right\}$$

Analogous to $\langle u_1, \dots, u_m \rangle$ but $\otimes \rightarrow \oplus$
 $|C| = 2^m$

→ Parity Checks

Minimal list of string $v_1, \dots, v_r \in \Sigma^n$ such that

$$C = \{u \in \Sigma^n : u \cdot v_1 = \dots = u \cdot v_r = 0\}$$

$$|C| = 2^{n-r} \quad \rightarrow \text{Each } v \rightarrow \frac{1}{2} \text{ space}$$

$$|C| = 2^m = 2^{n-r} \Rightarrow n = m + r$$

→ Ex. → 3-bit repetition code



→ (7, 4, 3) - Hamming code



Parity checks are equivalent to $5+6$. gen. containing only Z, I

→ Ex. → 3-bit repetition code



→ $(7, 4, 3)$ - Hamming Code

